Centurion University
Shaping Lives... Empowering Communities...

School: ..... Campus: .....

Academic Year: ..... Subject Name: ..... Subject Code: .....

Semester: ..... Program: ..... Branch: ..... Specialization: .....

Date: .....

Applied and Action Learning
(Learning by Doing and Discovery)

Name of the Experiment:- Apply data transformation techniques such as normalization and standardization.

Coding Phase:-Pseudo Code / Flow Chart / Algorithm

BEGIN

Step 1: Import Required Libraries
- **Import pandas**
- **Import numpy**
- **Import seaborn**
- **Import matplotlib**
- **Import MinMaxScaler**
- **Import StandardScaler**

Step 2: Create the Dataset
- **Create a dictionary containing:**
  - **Name**
  - **Age**
  - **Gender**
  - **Marks**
  - **City**
- **Convert the dictionary into a DataFrame**

Step 3: Perform Normalization
- **Initialize MinMaxScaler**
- **Select numerical columns:**
  - **Age**
  - **Marks**
- **Apply Min-Max Normalization**
- **Store the normalized values in the DataFrame**
- **Display the normalized dataset**

Step 4: Perform Standardization
- **Recreate the original DataFrame**
- **Replace missing values in Age with the mean**
- **Replace missing values in Marks with the mean**
- **Initialize StandardScaler**
- **Select numerical columns:**
  - **Age**
  - **Marks**
- **Apply Standardization**
- **Store the standardized values in the DataFrame**
- **Display the standardized dataset**

END

Testing Phase:-Compilation of Code (error detection)

```
  Cell In[46], line 4

    ^
_IncompleteInputError: incomplete input
```

Fix Code

Implementation Phase :- Final Output(no error)

Step 1: Import Required Libraries

```
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Create / Load Dataset

```
data = {
  "Name": [
    "Amit", "Priya", "Rahul", "Sneha", "Rohan",
    "Anjali", "Vikash", "Neha", "Suman", "Karan",
    "Pooja", "Arjun", "Meena", "Rakesh", "Nisha",
    "Deepak", "Kavita", "Ajay", "Simran", "Ravi"
  ],

  "Age": [
    25, 30, np.nan, 28, 35,
    22, 29, np.nan, 31, 27,
    26, 34, 24, 32, np.nan,
    29, 23, 36, 28, 30
  ],

  "Gender": [
    "Male", "Female", "Male", "Female", "Male",
    "Female", "Male", "Female", "Female", "Male",
    "Female", "Male", "Female", "Male", "Female",
    "Male", "Female", "Male", "Female", "Male"
  ],

  "Salary": [
    40000, 50000, 45000, np.nan, 60000,
    38000, 52000, 48000, np.nan, 47000,
    43000, 62000, 39000, 58000, 45000,
    np.nan, 41000, 65000, 49000, 53000
  ],

  "Department": [
    "HR", "IT", "Sales", "Finance", "IT",
    "HR", "Sales", "Finance", "IT", "HR",
    "Sales", "IT", "Finance", "HR", "Sales",
    "IT", "Finance", "HR", "Sales", "IT"
  ],
```

```
    "Experience": [
        2, 5, 3, 4, np.nan,
        1, 6, 2, 5, 3,
        np.nan, 7, 2, 6, 4,
        5, 1, 8, np.nan, 4
    ]
}

# Convert into DataFrame
df = pd.DataFrame(data)
```

Step 3: Normalization

```
# Initialize MinMaxScaler
scaler = MinMaxScaler()
# Select Numerical Columns
num_col = ["Age", "Salary", "Experience"]
# Apply Normalization
df[num_col] = scaler.fit_transform(df[num_col])
# Display Dataset
print(df.head())
```

```
    Name      Age  Gender    Salary Department  Experience
0   Amit  0.214286    Male  0.074074         HR    0.142857
1  Priya  0.571429  Female  0.444444         IT    0.571429
2  Rahul       NaN    Male  0.259259      Sales    0.285714
3  Sneha  0.428571  Female       NaN    Finance    0.428571
4  Rohan  0.928571    Male  0.814815         IT         NaN
```

Step 4: Standardization

```
df = pd.DataFrame(data)
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
df["Experience"] = df["Experience"].fillna(df["Experience"].mean())
#Initialize StandardScaler
scaler = StandardScaler()
num_col = ["Age", "Salary", "Experience"]
# Apply Standardization
df[num_col] = scaler.fit_transform(df[num_col])
# Display Dataset
print(df.head())
```

```
    Name          Age  Gender    Salary Department  Experience
0   Amit -1.034129e+00    Male -1.234051         HR   -1.084652
1  Priya  3.393237e-01  Female  0.119424         IT    0.542326
2  Rahul  9.758972e-16    Male -0.557314      Sales   -0.542326
3  Sneha -2.100575e-01  Female  0.000000    Finance    0.000000
4  Rohan  1.712777e+00    Male  1.472900         IT    0.000000
```

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	**50**		

Signature of the Student:

Name :

Regn. No. :

Signature of the Faculty: